Secure cPanel & OpenVPN on AWS

Production-Grade Terraform Infrastructure

VPN-Only Admin Access | Split Tunnel | Encrypted Backups | Full Monitoring

7	2	8	S
7 Modules	2 Environments	8 Test Files	State Locking

Architecture Overview

VPC with public/private subnets, VPN-only admin, public web hosting

VPC 10.0.0.0/16 [Flow Logs → CloudWatch]

Public Subnet (NACL protected)

OpenVPN Server

t3.small • Ubuntu 22.04 • EIP
UDP 1194 • Split Tunnel VPN
Easy-RSA PKI • AES-256-GCM
PKI backup → S3 • VPC DNS push
IMDSv2 required • Encrypted EBS

VPN tunnel

Private Subnet (NACL protected)

cPanel/WHM Server

c5.xlarge • AlmaLinux 8
gp3 100GB (3000 IOPS, 125 MiB/s)
HTTP/HTTPS: public (hosted sites)
WHM/cPanel/SSH: VPN-only
IAM role for S3 backups
IMDSv2 required • Encrypted EBS

Internet GW
Public traffic

NAT Gateway
Private → internet

S3 VPC Endpoint
Free, no NAT needed

VPN Return Route
Private RT → VPN ENI

VPN Clients (10.8.0.0/24)

Public Web (80/443)

Security Design

Defense in depth: network, access, encryption, and audit controls

V — VPN-Only Admin Access

- WHM (2087) — VPN CIDR only
- cPanel (2083) — VPN CIDR only
- SSH (22) — VPN CIDR only
- admin_cidr rejects 0.0.0.0/0

P — Public Web Hosting

- HTTP (80) — 0.0.0.0/0
- HTTPS (443) — 0.0.0.0/0
- For hosted websites only
- Separate from admin ports

S — Split Tunnel VPN

- Only private subnets via VPN
- Internet traffic stays direct
- VPC DNS pushed to clients
- VPN return route in private RT

N — Network Controls

- Public & Private NACLs
- VPC Flow Logs → CloudWatch
- Source/dest check disabled (VPN)
- Security Groups per role

E — Encryption & Storage

- EBS gp3 encrypted at rest
- S3 KMS server-side encryption
- S3 SSL-only bucket policy
- S3 VPC Gateway Endpoint

I — Instance Hardening

- IMDSv2 required (both instances)
- IAM roles — no static keys
- PKI auto-backup to S3
- DynamoDB state locking

Backup & Monitoring

Automated S3 backups with encryption + CloudWatch alarms with SNS alerts

S3 Backup Infrastructure

Versioned S3 Bucket	Full version history for recovery
KMS Encryption	Server-side encryption with AWS managed key
SSL-Only Policy	Bucket policy denies non-HTTPS requests
Public Access Block	All public access blocked at bucket level
Lifecycle Rules	Automatic cleanup of old versions
OpenVPN PKI Backup	Certificates auto-backed up on first boot
cPanel IAM Role	Instance role for backup — no static keys
S3 VPC Endpoint	Direct S3 access, no NAT charges

CloudWatch Monitoring + SNS

CPU Utilization
> 80% for 5 min

Status Check Failed
Instance or system failure

Disk Utilization
> 85% (CWAgent)

Memory Utilization
> 85% (CWAgent)

SNS Email Alerts
Immediate notification

VPC Flow Logs
All traffic → CloudWatch

Environment Comparison

Dev and Prod environments with consistent security, different sizing

Component	Dev	Prod
cPanel Instance	c5.large	c5.xlarge
OpenVPN Instance	t3.small	t3.small
Root Volume	100 GB gp3	100 GB gp3
Volume IOPS	3000	3000
Volume Throughput	125 MiB/s	125 MiB/s
Monitoring	CPU + Status + Disk + Memory	CPU + Status + Disk + Memory
S3 Backup	Versioned + KMS + SSL-only	Versioned + KMS + SSL-only
VPN	Split tunnel, PKI → S3	Split tunnel, PKI → S3
Flow Logs	Enabled	Enabled
State Locking	DynamoDB	DynamoDB
IMDSv2	Required	Required

Testing & Project Structure

Native Terraform tests with mock providers — no real resources created

Test Coverage (8 files)

vpc.tftest.hcl	VPC CIDR, subnets, Flow Logs, S3 Endpoint
security_groups.tftest.hcl	Admin VPN-only, web public, SSH restricted
openvpn.tftest.hcl	Instance type, EIP, source_dest_check, IAM
cpanel.tftest.hcl	Instance, volume, IOPS, IMDSv2, IAM profile
s3_backup.tftest.hcl	Versioning, encryption, public block, SSL-only
monitoring.tftest.hcl	CPU, status, disk, memory alarms + SNS
integration.tftest.hcl	Cross-module security model validation
admin_cidr.tftest.hcl	Rejects 0.0.0.0/0 for admin access

Project Structure

```
vcode/
├── modules/
│   ├── vpc/              # VPC, subnets, IGW, NAT, NACLs,
│   │                       Flow Logs, S3 Endpoint
│   ├── security_groups/  # SGs: admin VPN-only, web public
│   ├── openvpn/          # EC2, EIP, split tunnel, PKI→S3
│   ├── cpanel/           # EC2, IAM, hostname, gp3 tuning
│   ├── s3_backup/        # Encrypted, versioned, SSL-only
│   ├── route53/          # DNS A records
│   └── monitoring/       # CPU, status, disk, memory + SNS
├── environments/
│   ├── dev/              # c5.large, DynamoDB locking
│   └── prod/             # c5.xlarge, DynamoDB locking
├── tests/                # 8 native Terraform test files
└── README.md
```